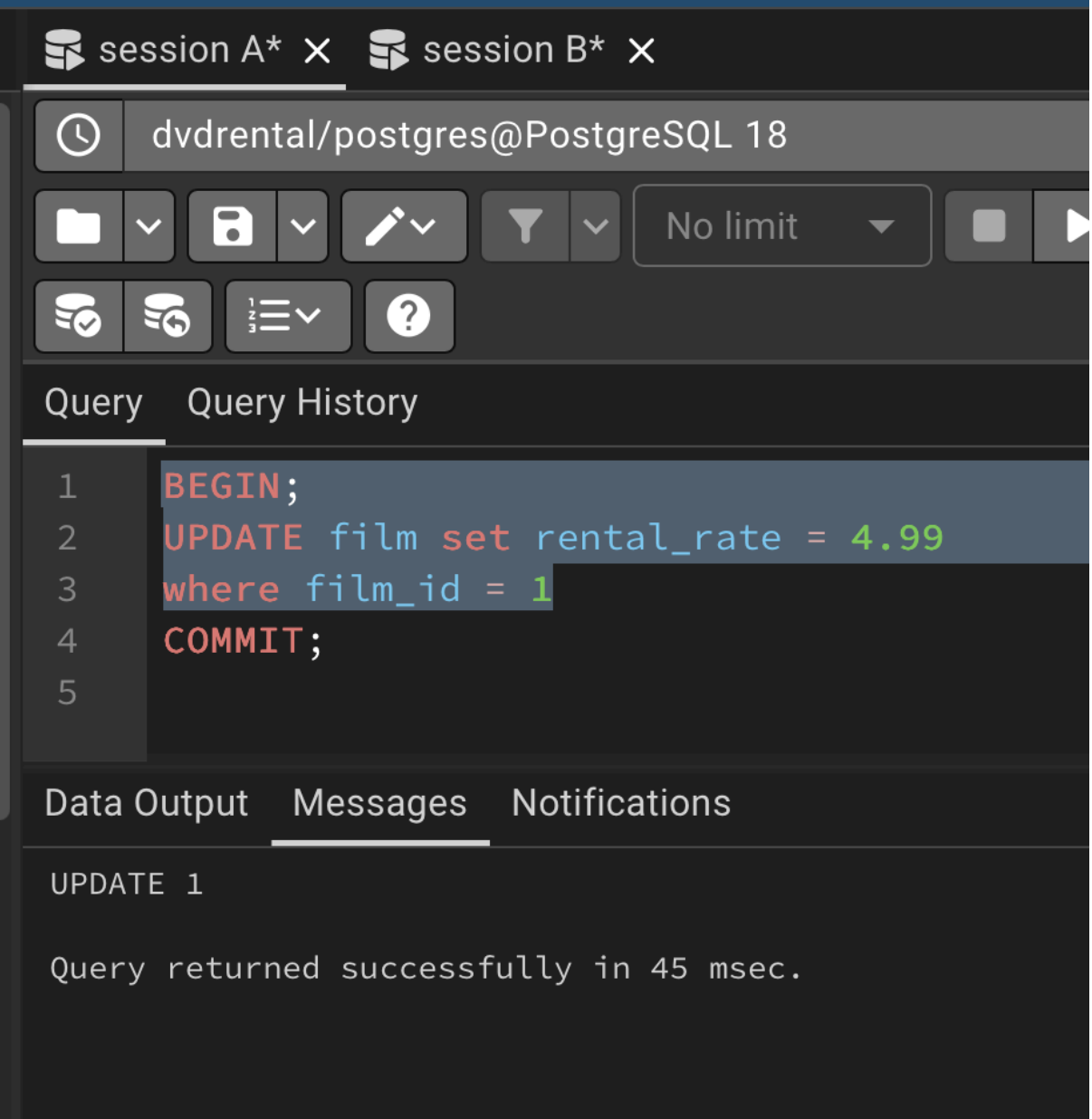


name: Bùi Hữu Gia Bảo
class: DS66B
id: 11247144

LAB 1:

STEP 1:



The screenshot shows a PostgreSQL client interface with two sessions, 'session A*' and 'session B*', open. The active session is 'dvdrental/postgres@PostgreSQL 18'. The interface includes a toolbar with icons for file operations, a filter dropdown set to 'No limit', and a query editor. The query editor displays a four-line SQL statement: `BEGIN;`, `UPDATE film set rental_rate = 4.99`, `where film_id = 1`, and `COMMIT;`. Below the query editor, the 'Data Output' tab is selected, showing the result 'UPDATE 1' and a message: 'Query returned successfully in 45 msec.'

```
BEGIN;  
UPDATE film set rental_rate = 4.99  
where film_id = 1  
COMMIT;
```

UPDATE 1

Query returned successfully in 45 msec.

STEP 2:

```
1 BEGIN;
2 UPDATE film set rental_rate = 4.99
3 where film_id = 1
4 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 51 msec.

Total rows:	Query complete 00:00:00.051
-------------	-----------------------------

STEP 3:

```
1 BEGIN;  
2 UPDATE film set rental_rate = 4.99  
3 where film_id = 1  
4 COMMIT;  
5
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 51 msec.

Total rows:	Query complete 00:00:00.051
-------------	-----------------------------

STEP 4:

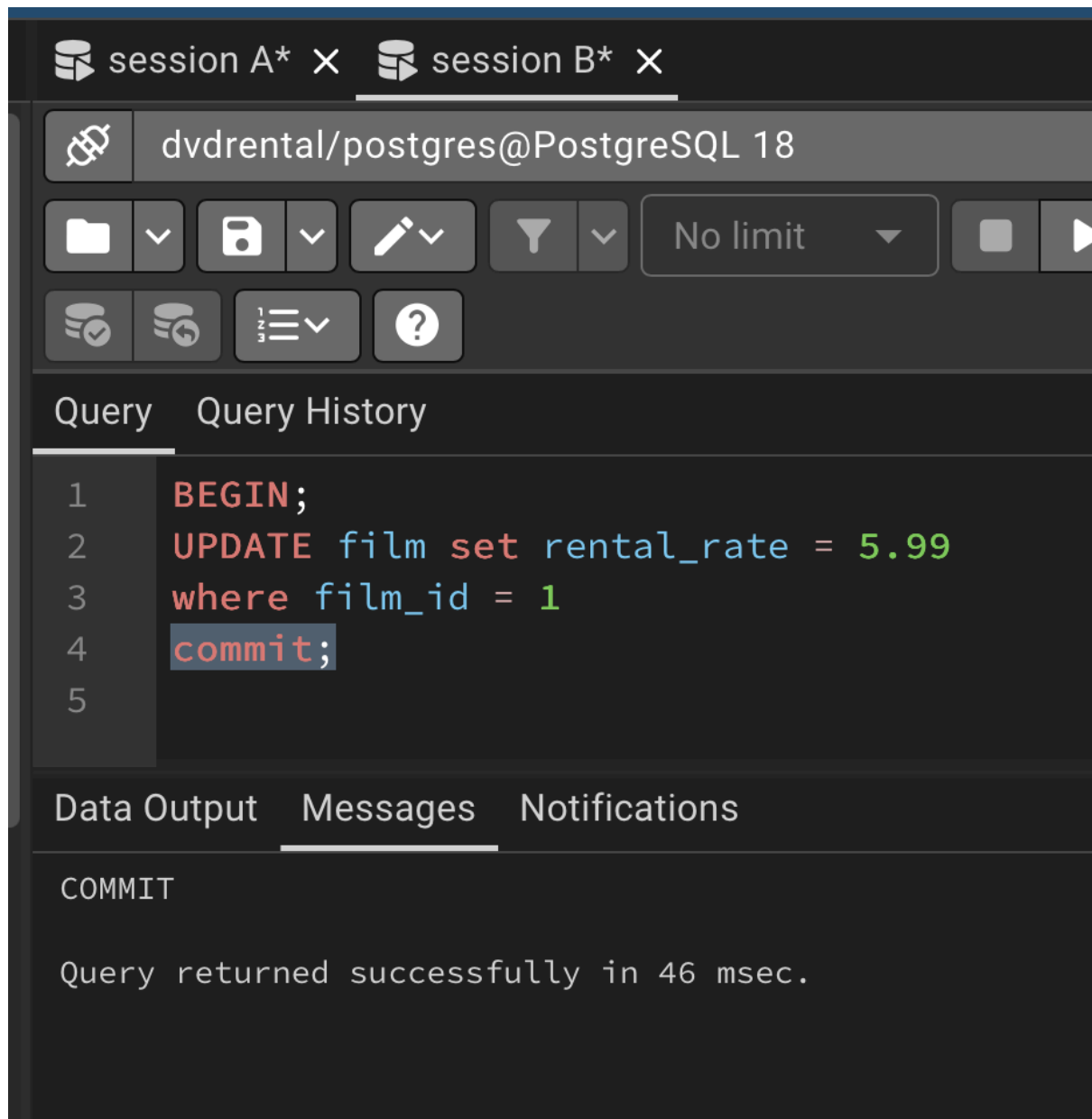
```
1 BEGIN;  
2 UPDATE film set rental_rate = 5.99  
3 where film_id = 1  
4 commit;  
5
```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 11 secs 33 msec.

Total rows:	Query complete 00:00:11.033
-------------	-----------------------------



LAB 2:
STEP 1:

session A* X session B X

dvdrental/postgres@PostgreSQL 18

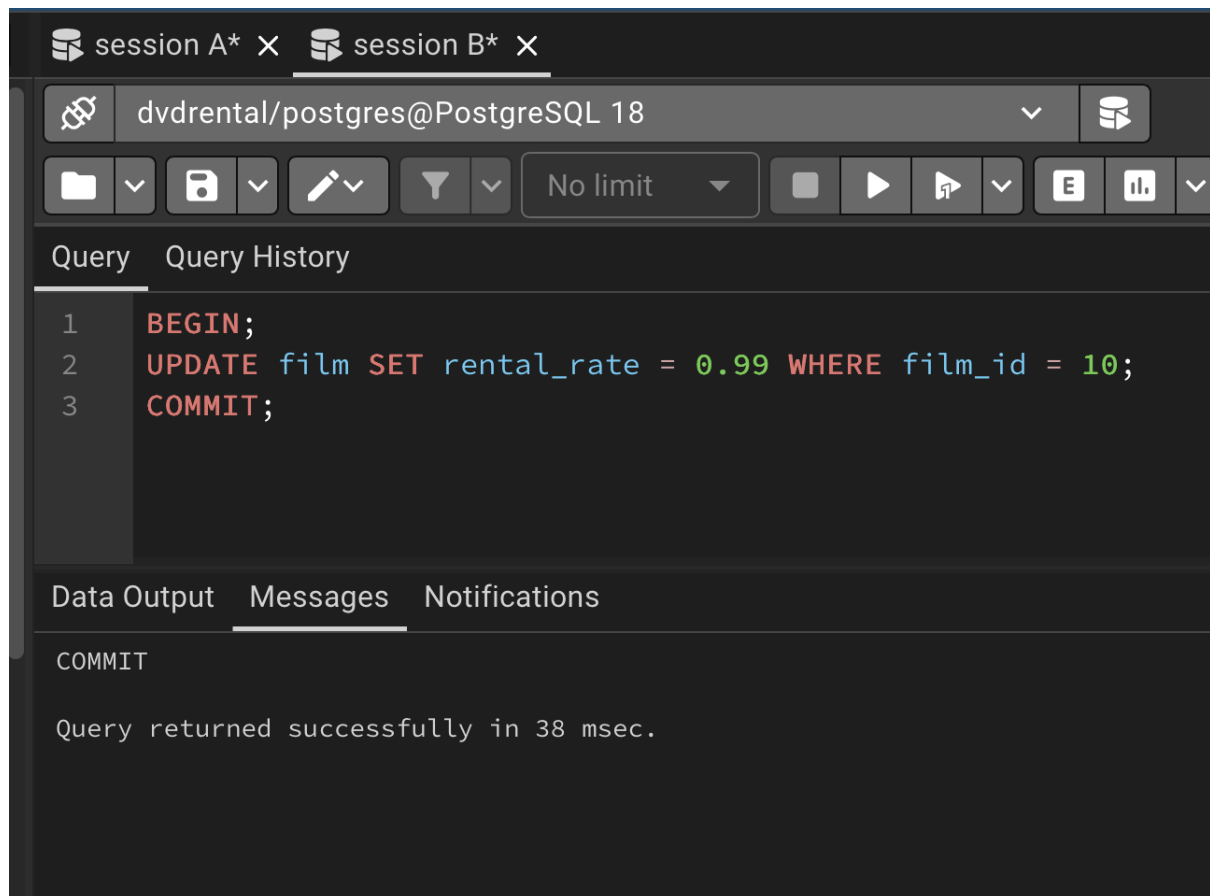
Query Query History

```
1 BEGIN;  
2 SELECT rental_rate FROM film WHERE film_id = 10;
```

Data Output Messages Notifications

	rental_rate numeric (4,2) 🔒
1	4.99

STEP 2:



STEP 3:

session A* X session B* X

dvdrental/postgres@PostgreSQL 18

Query Query History

```
1 BEGIN;  
2 SELECT rental_rate FROM film WHERE film_id = 10;  
3 SELECT rental_rate FROM film WHERE film_id = 10;  
4 COMMIT;
```

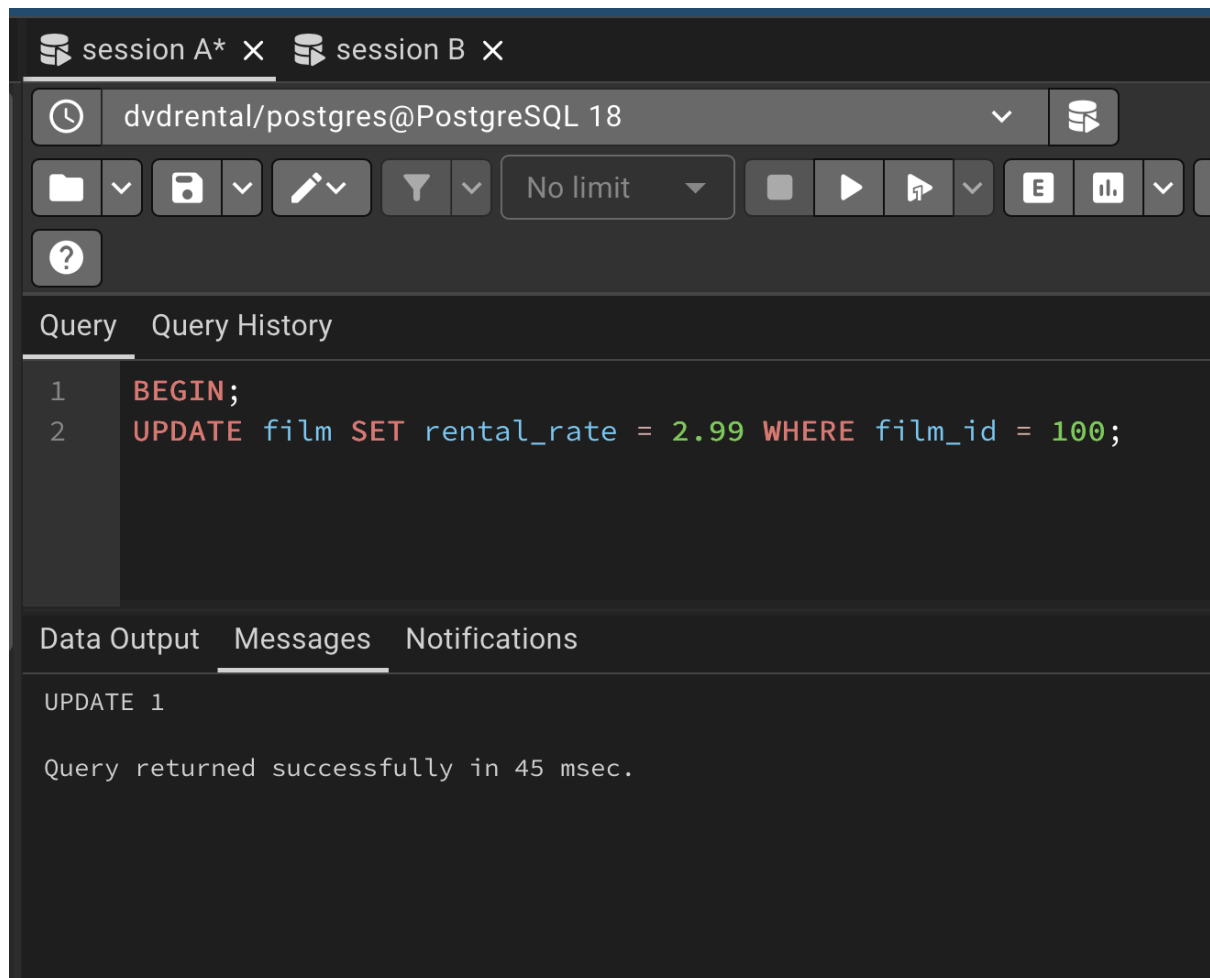
Data Output Messages Notifications

Showing rows: 1 to 1 of 1

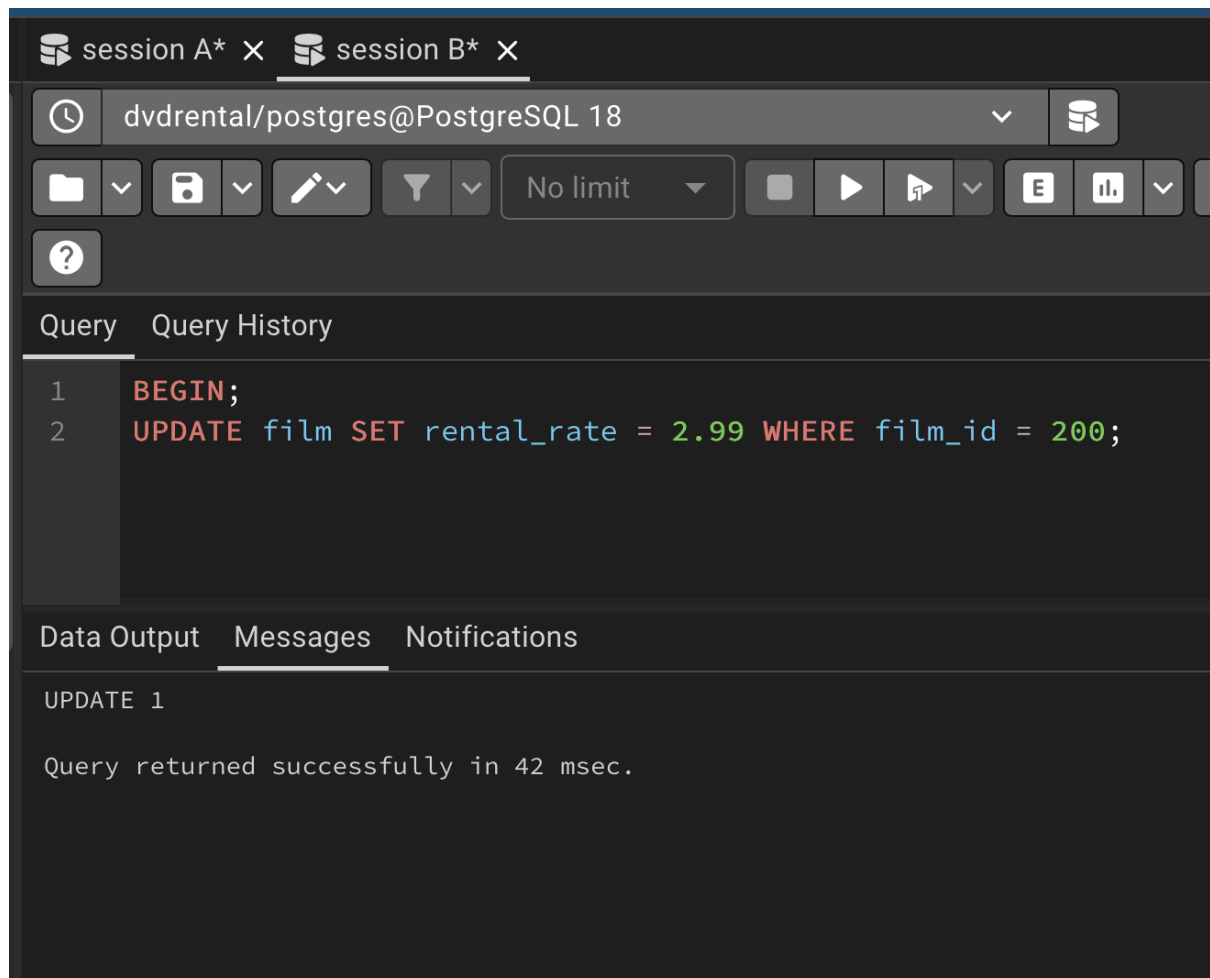
	rental_rate numeric (4,2)
1	0.99

LAB 3:

STEP 1



STEP 2:



STEP 3:

session A* X session B* X

dvdrental/postgres@PostgreSQL 18

Query Query History

```
1 BEGIN;  
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 100;  
3  
4 UPDATE film SET rental_rate = 3.99 WHERE film_id = 200;
```

Data Output Messages Notifications

Total rows: Waiting for the query to complete... 00:00:06.560

STEP 4:

session A* × session B* ×

! dvdrental/postgres@PostgreSQL 18

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Query Query History

1

BEGIN;

2

UPDATE film SET rental_rate = 2.99 WHERE film_id = 200;

3

4

UPDATE film SET rental_rate = 3.99 WHERE film_id = 100;

Data Output Messages Notifications

ERROR: deadlock detected
Process 32505 waits for ShareLock on transaction 1011; blocked by process 32500.
Process 32500 waits for ShareLock on transaction 1012; blocked by process 32505.

SQL state: 40P01
Detail: Process 32505 waits for ShareLock on transaction 1011; blocked by process 32500.
Process 32500 waits for ShareLock on transaction 1012; blocked by process 32505.
Hint: See server log for query details.
Context: while locking tuple (5,12) in relation "film"

Total rows: Query complete 00:00:01.102 LF Ln 4, Co

The screenshot shows a PostgreSQL query editor interface. At the top, there are tabs for 'session A*' and 'session B*'. Below the tabs, the connection string 'dvdrental/postgres@PostgreSQL 18' is displayed. A toolbar with various icons for file operations, query execution, and settings is visible. The main area is divided into two tabs: 'Query' and 'Query History'. The 'Query' tab is active, showing a SQL script with four lines: 1. `BEGIN;`, 2. `UPDATE film SET rental_rate = 2.99 WHERE film_id = 100;`, 3. (empty line), 4. `UPDATE film SET rental_rate = 3.99 WHERE film_id = 200;`. Below the query editor, there are three tabs: 'Data Output', 'Messages', and 'Notifications'. The 'Messages' tab is active, displaying the text 'UPDATE 1' and 'Query returned successfully in 26 secs 787 msec.'. At the bottom of the interface, a status bar shows 'Total rows: Query complete 00:00:26.787' and 'Ln 4, Co'.

```
1 BEGIN;
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 100;
3
4 UPDATE film SET rental_rate = 3.99 WHERE film_id = 200;
```

UPDATE 1

Query returned successfully in 26 secs 787 msec.

Total rows: Query complete 00:00:26.787 LF Ln 4, Co

LAB 4:

STEP 1:

session A* X session B X

dvdrental/postgres@PostgreSQL 18

Query Query History

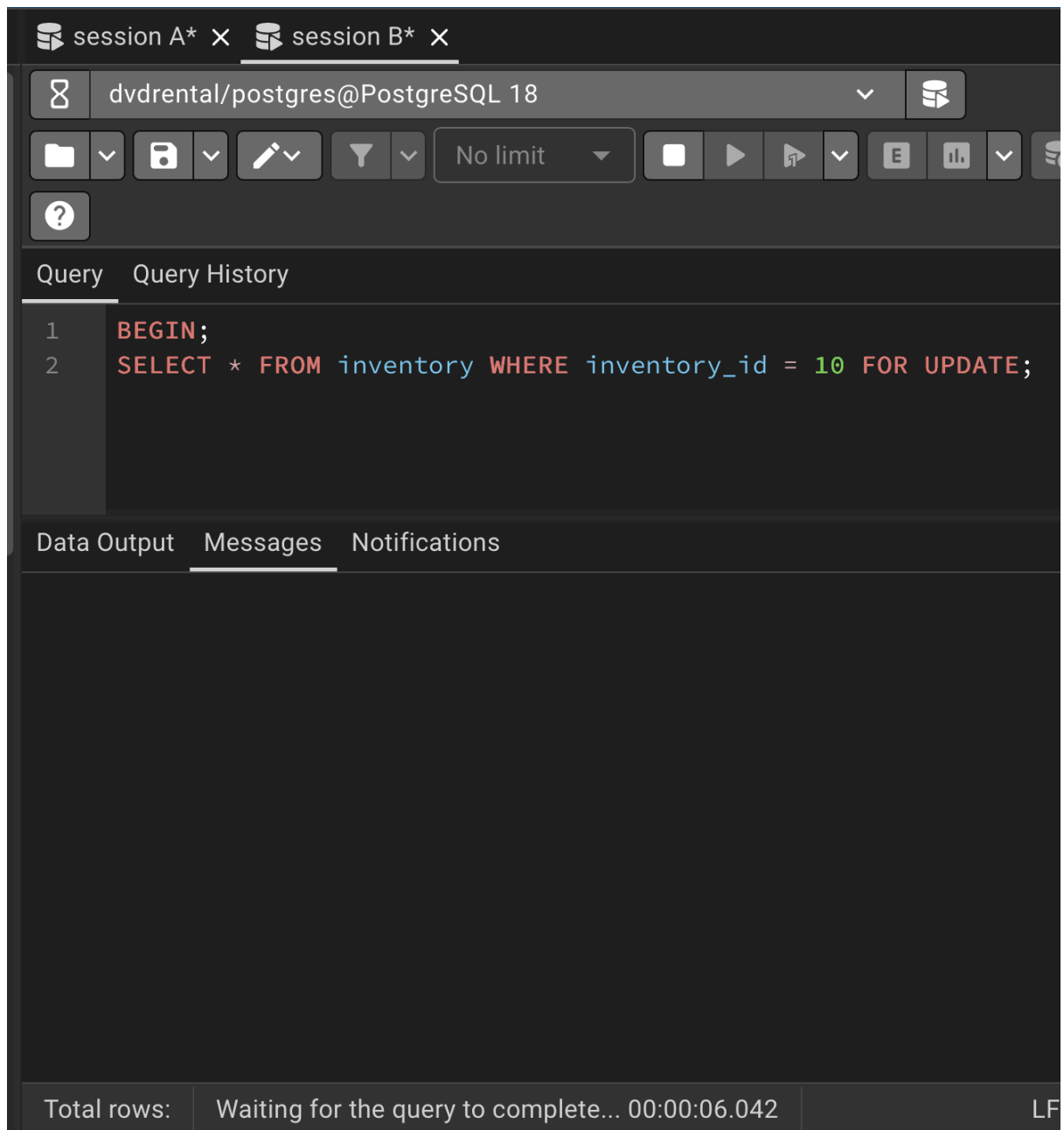
```
1 BEGIN;  
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1 Page No: 1

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

STEP 2:



STEP 3:

The screenshot shows a PostgreSQL client window with two tabs: 'session A*' and 'session B*'. The active session is 'session B*'. The connection string is 'dvdrental/postgres@PostgreSQL 18'. The interface includes a toolbar with icons for file operations, query execution, and settings. The 'Query' tab is selected, displaying a SQL script with six lines. The script is as follows:

```
1 BEGIN;  
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;  
3  
4 INSERT INTO rental (rental_date, inventory_id, customer_id, staff_id)  
5 VALUES (NOW(), 10, 1, 1);  
6 COMMIT;
```

The 'Messages' tab is also visible, showing an error message:

```
ERROR: column "is_rented" of relation "inventory" does not exist  
LINE 2:     SET is_rented = TRUE  
          ^  
QUERY: UPDATE inventory  
        SET is_rented = TRUE  
        WHERE inventory_id = NEW.inventory_id  
CONTEXT: PL/pgSQL function trg_check_inventory_func() line 3 at SQL statement  
  
SQL state: 42703
```

STEP 4:

session A* × session B* ×

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dvdrental/postgres@PostgreSQL 18

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Query Query History

1

BEGIN;

2

SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;

Data Output Messages Notifications

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SQL

Showing rows: 1 to 1 ✎ 🖨️ Page No: 1 of 1 ⏪ ⏩ ⏴ ⏵

	inventory_id [PK] integer ✎	film_id smallint ✎	store_id smallint ✎	last_update timestamp without time zone ✎
1	10	2	2	2006-02-15 10:09:17